

IoT Challenge #3 – Part 2: Exercise

LoRaWAN – Spreading Factor Analysis

Anasuya Satapathy
Golsa Shokri

May 2026

Problem Statement

A LoRaWAN network in Europe operates at carrier frequency **868 MHz** with **bandwidth 125 kHz**. The network is composed of **one gateway** and **40 sensor nodes**. Each sensor node transmits packets with **payload size 20 bytes** according to a Poisson process with intensity $\lambda = 2$ packets/minute. All nodes use the same spreading factor (SF).

EQ1 – Largest SF Guaranteeing Packet Success Rate $\geq 75\%$

Step 1: LoRaWAN Frame Size

In LoRaWAN the application payload of 20 bytes is wrapped in a MAC frame that adds a fixed overhead of **13 bytes**:

Field	Size
MHDR (MAC header)	1 byte
DevAddr	4 bytes
FCtrl	1 byte
FCnt (frame counter)	2 bytes
FPort	1 byte
MIC (message integrity code)	4 bytes
Total overhead	13 bytes

Therefore the total LoRa frame size fed to the radio is:

$$P_L = 20 + 13 = \mathbf{33} \text{ bytes}$$

All airtimes below were read from the **TTN airtime calculator** (<https://www.thethingsnetwork.org/airtime-calculator>) using: $P_L = 20$ bytes application payload (the calculator adds

the 13-byte overhead internally), BW = 125 kHz, coding rate 4/5, standard 8-symbol preamble, explicit header, CRC enabled.

Step 2: Airtime for Each Spreading Factor (SF7–SF12)

SF	Low-Rate Opt.	T_{air} from TTN calculator (ms)
7	No	71.9
8	No	133.6
9	No	246.8
10	No	452.6
11	Yes	987.1
12	Yes	1810.4

Table 1: Packet airtime obtained from the TTN airtime calculator (BW = 125 kHz, payload = 20 bytes application, EU868).

Note: the TTN calculator value for SF8 is ≈ 133 ms, which is used as the reference in the computation below.

Step 3: Pure ALOHA Collision Model

The problem states that transmissions are *uncoordinated and may overlap in time*, which corresponds to the **pure (unslotted) ALOHA** model.

In pure ALOHA with N nodes each generating traffic at rate λ (packets/s) and packet duration T_{air} , the *vulnerability window* of a packet spans $2T_{\text{air}}$. Because all other nodes follow independent Poisson processes, the probability that *no other packet* starts in that window is:

$$P_{\text{success}} = e^{-2G}$$

where the normalised offered load is:

$$G = N \cdot \lambda \cdot T_{\text{air}}$$

Network parameters:

$$N = 40, \quad \lambda = \frac{2}{60} \frac{\text{packets}}{\text{s}} = \frac{1}{30} \text{ s}^{-1}$$

$$G = 40 \times \frac{1}{30} \times T_{\text{air}} = \frac{4}{3} T_{\text{air}} \quad (T_{\text{air}} \text{ in seconds})$$

Step 4: Maximum Allowable Airtime

Setting $P_{\text{success}} \geq 0.75$:

$$\begin{aligned} e^{-2G} &\geq 0.75 \\ -2G &\geq \ln(0.75) \\ G &\leq \frac{-\ln(0.75)}{2} = \frac{0.2877}{2} = 0.1438 \end{aligned}$$

$$\frac{4}{3} T_{\text{air}} \leq 0.1438 \implies T_{\text{air}} \leq \frac{0.2877 \times 3}{2 \times 4} = 0.1079 \text{ s} = \mathbf{107.9 \text{ ms}}$$

Any spreading factor whose airtime is $\leq 107.9 \text{ ms}$ satisfies the requirement.

Step 5: Success Rate for Each SF

SF	T_{air} (ms)	$G = \frac{4}{3}T_{\text{air}}$	$P_{\text{success}} = e^{-2G}$	$\geq 75\%$?
7	71.9	0.0959	82.54%	✓
8	133.6	0.1782	70.02%	×
9	246.8	0.3290	51.78%	×
10	452.6	0.6035	29.91%	×
11	987.1	1.3162	7.19%	×
12	1810.4	2.4139	0.80%	×

Table 2: Offered load and packet success rate under pure ALOHA (40 nodes, $\lambda = 2$ packets/min, BW = 125 kHz).

Verification for SF7:

$$G = \frac{4}{3} \times 0.07194 = 0.0959 \quad P_{\text{success}} = e^{-2 \times 0.0959} = e^{-0.1918} \approx 82.5\% \geq 75\% \checkmark$$

Verification for SF8:

$$G = \frac{4}{3} \times 0.13363 = 0.1782 \quad P_{\text{success}} = e^{-2 \times 0.1782} = e^{-0.3563} \approx 70.0\% < 75\% \times$$

Answer to EQ1

The **largest spreading factor** that guarantees a packet success rate of at least 75% is **SF = 7** (success rate $\approx 82.5\%$).

SF8 ($T_{\text{air}} \approx 133 \text{ ms}$ from the TTN calculator) already drops the success rate to $\approx 70\%$, which is below the 75% threshold. All higher SFs perform even worse.

EQ2 – Diagnosing Non-Uniform Performance in the Field

Observation: After deployment, the measured packet success rate is *significantly lower than expected*, and performance is *non-uniform across nodes*: some nodes achieve good delivery rates while others experience very poor performance.

The most appropriate action is:

Option 2 – Move the nodes closer to the gateway

Justification

The critical clue is the **non-uniform** distribution of performance. Under the theoretical model in EQ1, all 40 nodes were assumed identical and equidistant, so every node experiences the same collision probability and the same expected success rate. In the real deployment, however, some nodes perform well while others perform very poorly. This asymmetry *cannot* be explained by collisions alone – collisions affect all nodes approximately equally when all nodes share the same SF and arrival rate.

The non-uniformity strongly points to **varying propagation distances**:

- **Nodes close to the gateway** receive strong signals (high RSSI/SNR), tolerate mild interference, and achieve good delivery rates.
- **Nodes far from the gateway** receive weak signals (RSSI near or below the sensitivity threshold), and suffer high packet loss even in the absence of collisions – yielding very poor delivery rates.

This is a classic **link-budget / coverage problem**. Moving the nodes closer to the gateway increases the received signal strength for all nodes, especially the distant ones, thus eliminating the root cause of non-uniformity.

Why Not the Other Options?

Option 1 – Decrease the number of nodes: Reducing N lowers the offered load G and raises the average success rate, but it does *not* cure non-uniformity. Distant nodes would still receive sub-threshold signals and continue to perform poorly regardless of how few neighbours they have.

Option 3 – Increase the spreading factor: A higher SF provides ≈ 3 dB more receiver sensitivity per SF step, which could help distant nodes. However, increasing SF *dramatically increases airtime* (e.g. SF8 is already 133 ms and drops success rate to 70%; SF12 reaches 1810 ms with a success rate below 1%). The resulting increase in collision probability harms all nodes globally and makes the overall network performance worse, not better.

Summary Table

Action	Effect	Verdict
Decrease nodes	Lowers collisions but does not fix poor SNR of distant nodes	Not appropriate
Move nodes closer	Improves received signal strength for all nodes; directly addresses the root cause of non-uniformity	Most appropriate
Increase SF	Slightly better sensitivity but greatly increases air-time and collisions	Counter-productive

Table 3: Comparison of candidate actions for EQ2.